Mechanical Properties I

↳ last topic on the midterm

Outline

- Introduction
- Mechanical testing
- Tensile test
- Elastic deformation

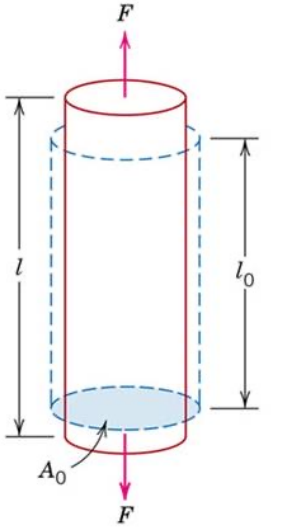
Mechanical Properties

- From an applications standpoint, one of the most important topics within Materials Sc. & Eng. is the study of how materials respond to external loading or deformation.
- Most components, even if used primarily for other property (e.g. electronic substrate) have to fulfil certain *mechanical functions* as well.
- Important mechanical properties are strength, hardness, stiffness, ductility and toughness.
- Laboratory testing to measure mechanical properties attempts to replicate the service conditions.
- Consistency is accomplished by using standardised test, so people are measuring same thing in the same way.
- **A**merican **S**ociety for **T**esting **M**aterials (ASTM) maintains and updates standards for mechanical properties.
- Several other standards organizations exist, e.g. SAE, ANSI, DIN….

Mechanical Testing of Metals

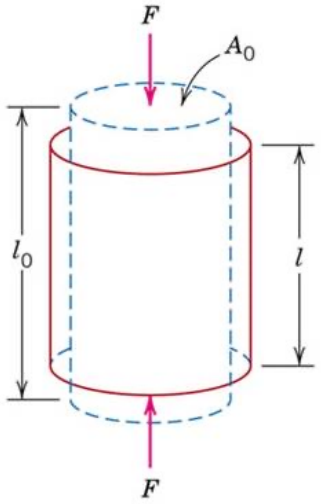
Loading can take any of the following forms:

tension



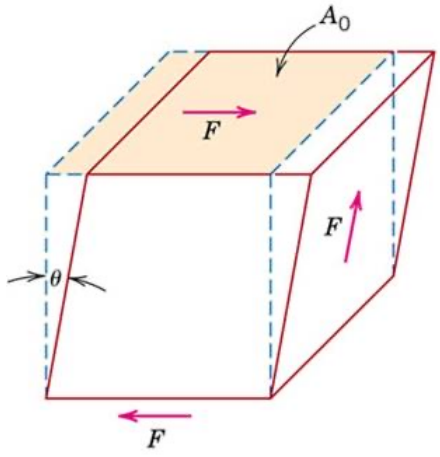
Pulling/ Stretching

compression



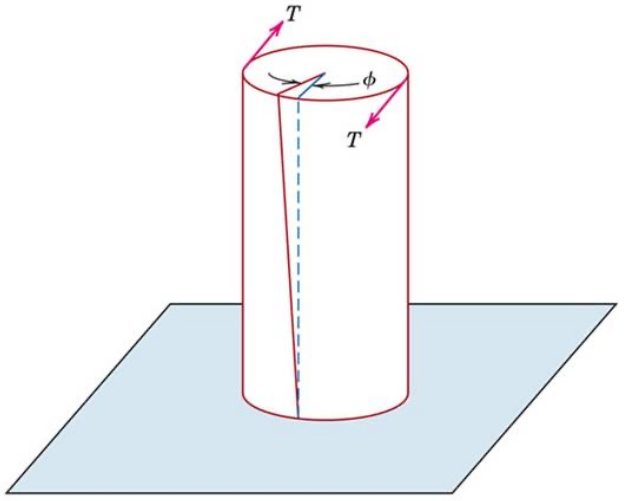
squeezing/ squashing

shear



sliding

torsion

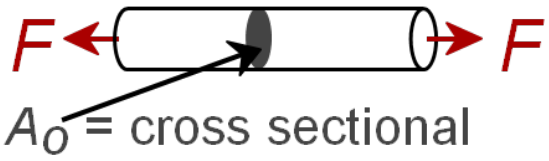


twisting

Different tests measure different types of loading conditions

Common States of Stress

Simple tension: cable



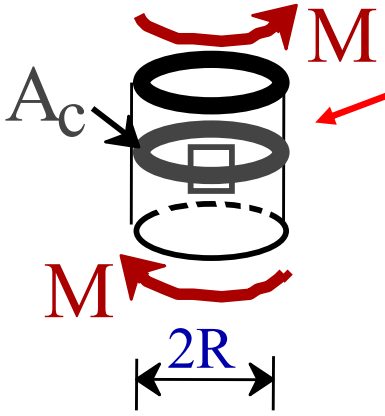
$$\sigma = \frac{F}{A_o}$$





Ski lift *(photo courtesy P.M. Anderson)*

Torsion: drive shaft



Simple compression:

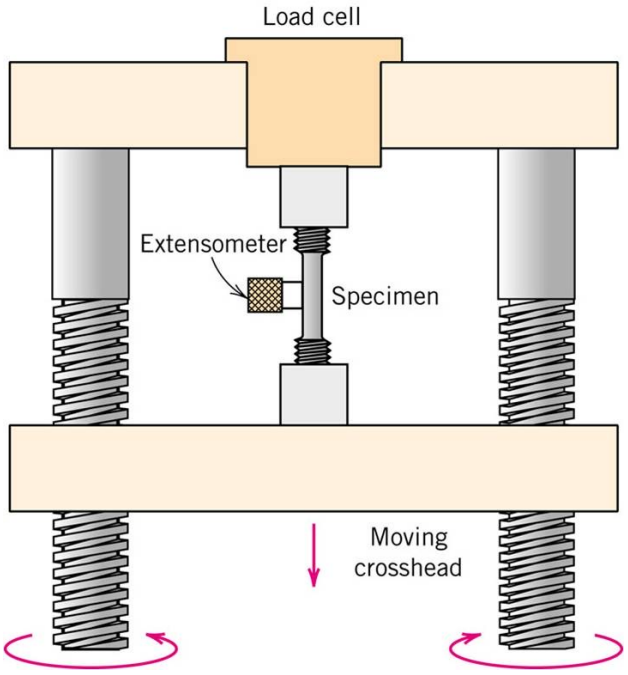




Canyon Bridge, Los Alamos, NM
(photo courtesy P.M. Anderson)

tries to stretch the sample and measures the force req to do so

Tensile Test



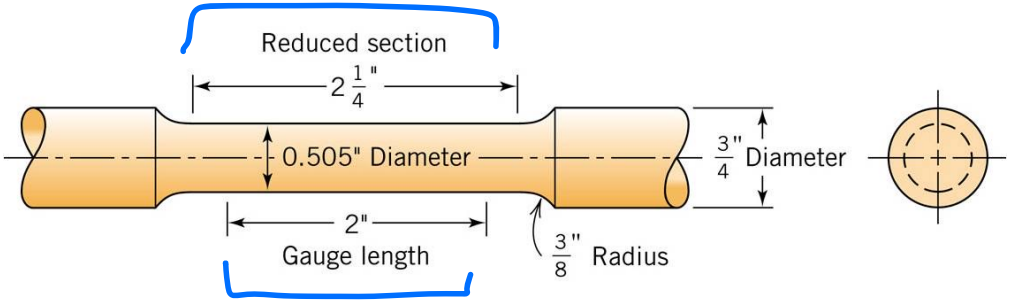
A typical tensile test machine

- The material's response to the applied tensile or compressive load is a change in length.
- We can monitor the change in length very precisely with an instrument called an *extensometer*.

$$\varepsilon = \frac{l_i - l_o}{l_o} = \frac{\Delta l}{l_o}$$

We call this quantity strain

ε = change in length / original length



A typical standard tensile test specimen

Where:

l = instantaneous length

l_o = initial length (length at various points during the test)

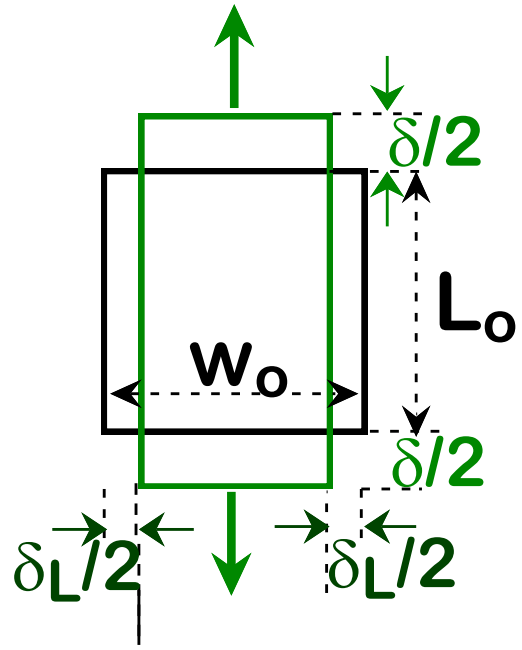
Strain is a dimensionless quantity (or, can be reported as m/m or in./in.)

Engineering Strain

- Tensile strain: $\varepsilon = \frac{\delta}{L_o}$

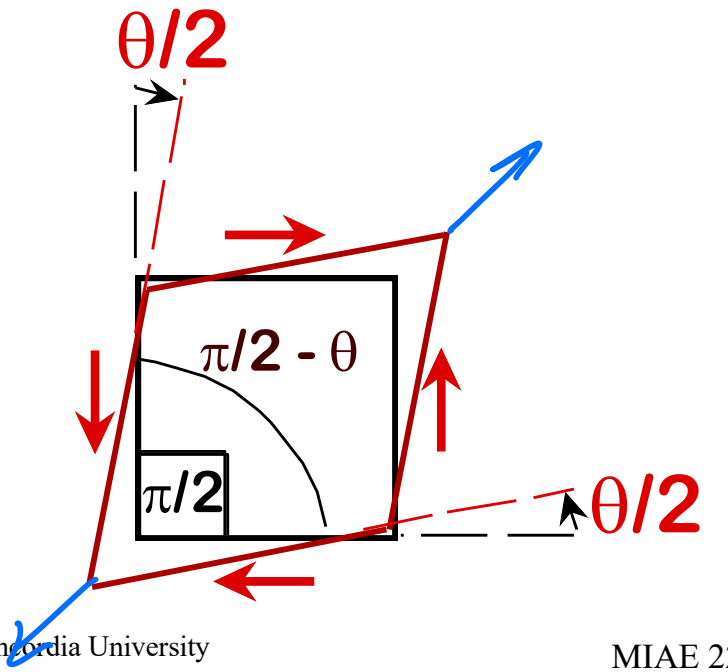
strain

- Lateral strain: $\varepsilon_L = \frac{-\delta_L}{w_o}$



- Shear strain:

$$\gamma = \tan\theta$$



Strain is always dimensionless.

Tensile Test

Typically, loading is normalized to cross sectional area:

$$\sigma = \frac{F}{A_o}$$

We refer to this ratio as the applied **stress**

when normalized to initial area, this is **engineering stress**.

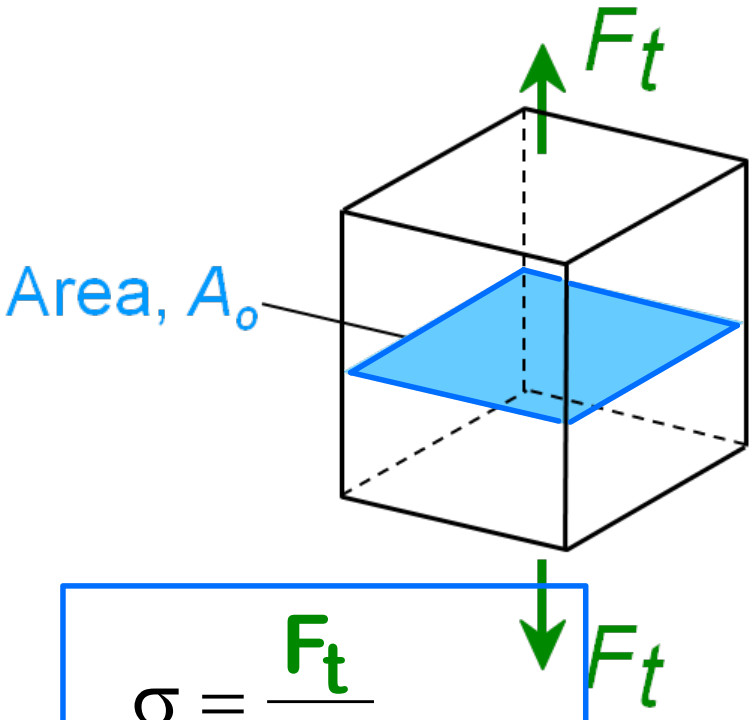
when normalized to actual area, this is **true stress**.

Tensile test

- Load - elongation testing
- But a thin wire breaks under a lower tensile load than a thicker wire of the same material
- Engineering stress (σ) versus engineering strain (ε)

Engineering Stress

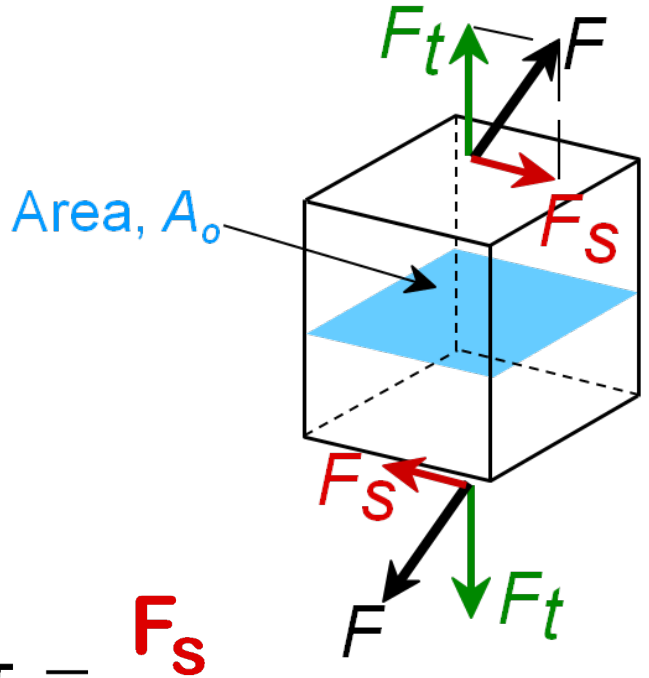
- Tensile stress, σ:



$$\sigma = \frac{F_t}{A_o}$$

A_o: original area before loading

- Shear stress, τ:

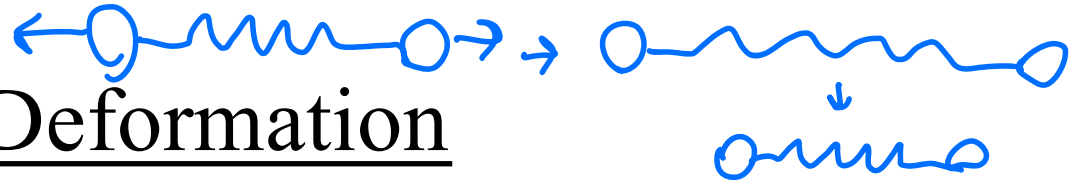


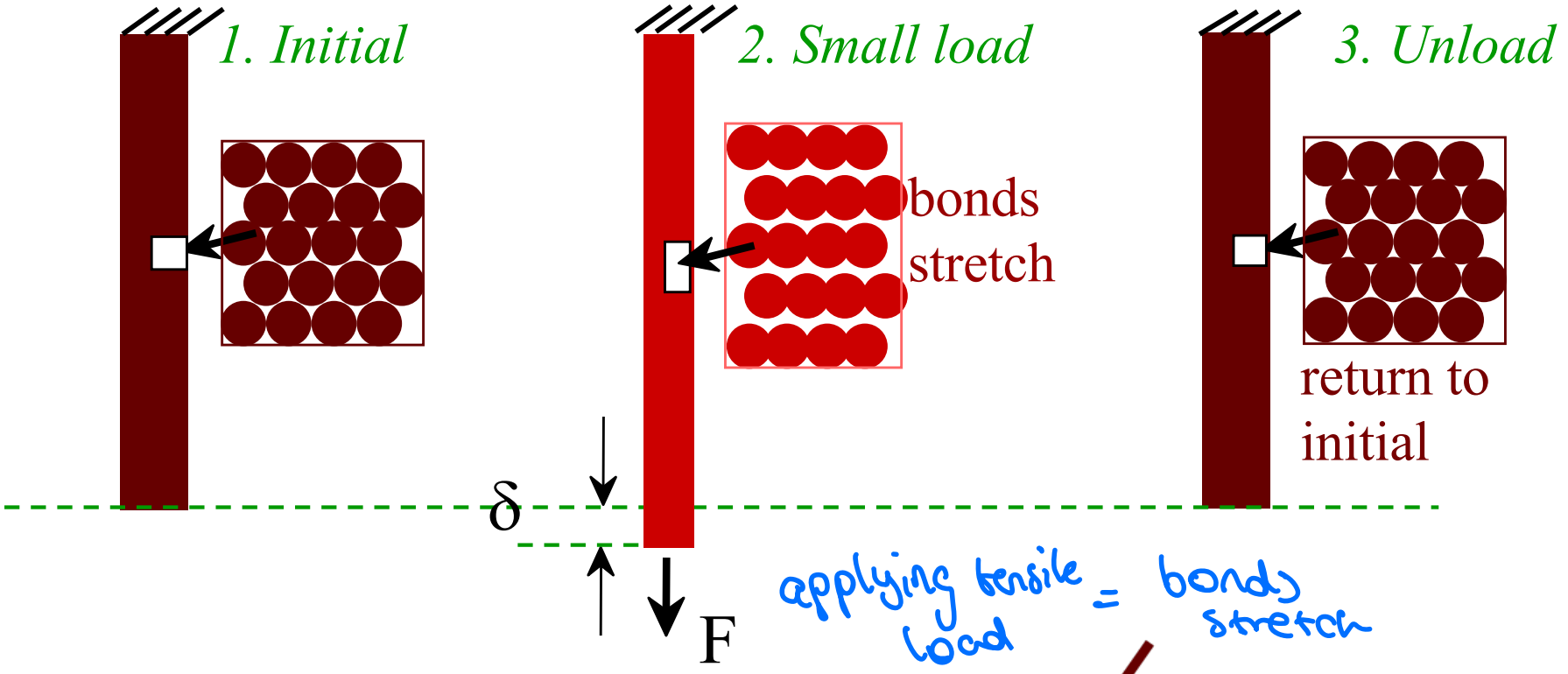
$$\tau = \frac{F_s}{A_o}$$

Shear stress

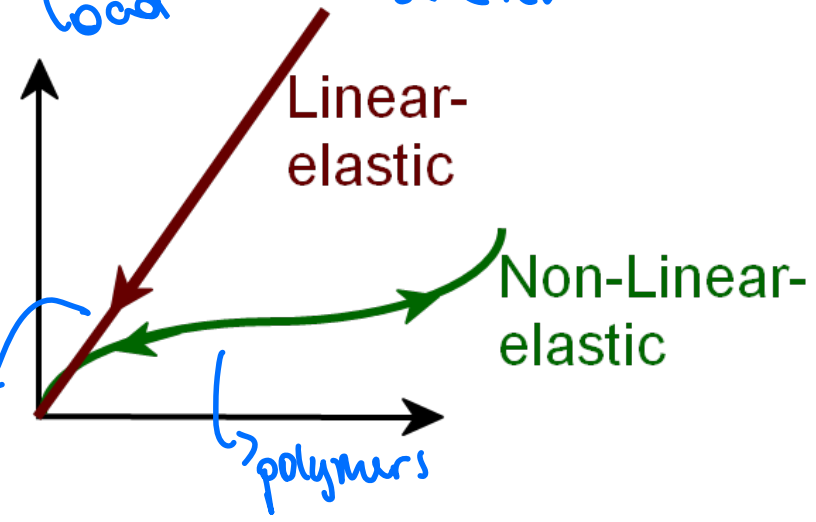
Stress has units: N/m^2 or lb/in^2

Elastic Deformation





applying tensile load = bonds stretch



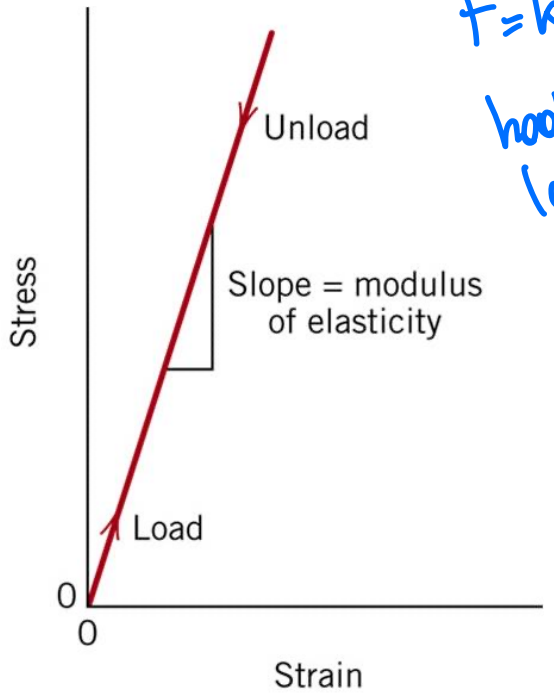
Elastic means reversible!

recoverable

Stress-Strain Behaviour - *Elastic Response*

Initially, stress and strain are directly proportional to each other

- *Rationale: atoms can be thought of as masses connected to each other through a network of* ***springs***.



$F = kx$

hooke's law

According to Hooke's law, the extension of a spring, x, and the applied force, F, are related by the spring constant, k:

$$F = - kx$$

$\sigma = E\varepsilon$ ↳ in terms of stress ↳ modulus of elasticity

The constant of proportionality, *Young's modulus or modulus of elasticity*, is a measure of the material's stiffness.

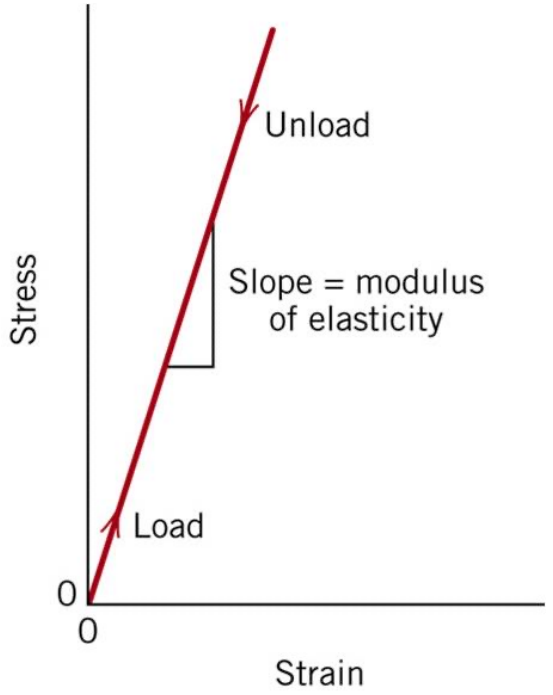
Materials possessing high stiffness: W, Ta, Mo → large... slope

Materials possessing low stiffness: Al, Cu, Ag → ..small... slope

Stress-Strain Behaviour - *Elastic Response*

In the elastic region a material returns to its original dimensions when load released, and we can write

$$\sigma = E\,\varepsilon \quad \text{or} \quad \frac{F}{A_o} = E\frac{\Delta l}{l_o}$$



Example:

A steel wire with a cross sectional area of 0.55 mm^2 and length of 10 m is extended *elastically* 1.68 mm by a force of 17.24 N. What is the modulus of elasticity for this steel specimen?

$A_o = 0.55\,mm^2$ $l_o = 10m$ $\Delta l = 1.68mm$

$F = 17.24N$ $E = ?$

$\rightarrow \sigma = \frac{F}{A_o} = \frac{17.24N}{0.55mm^2} = 31.34N/mm^2 = 31.34\times10^6\,\frac{N}{m^2}$

$\rightarrow \varepsilon = \frac{\Delta l}{l_o} = (1.68\times10^{-3}m)/(10m) = 0.000168$

$\rightarrow E = \sigma/\varepsilon = (31.34\times10^6 N/m^2)/$

Young's Modulus or Modulus of Elasticity

So E tells us how much something will stretch elastically when loaded, i.e. the **STIFFNESS** of that material

High E value - very stiff

Medium E value

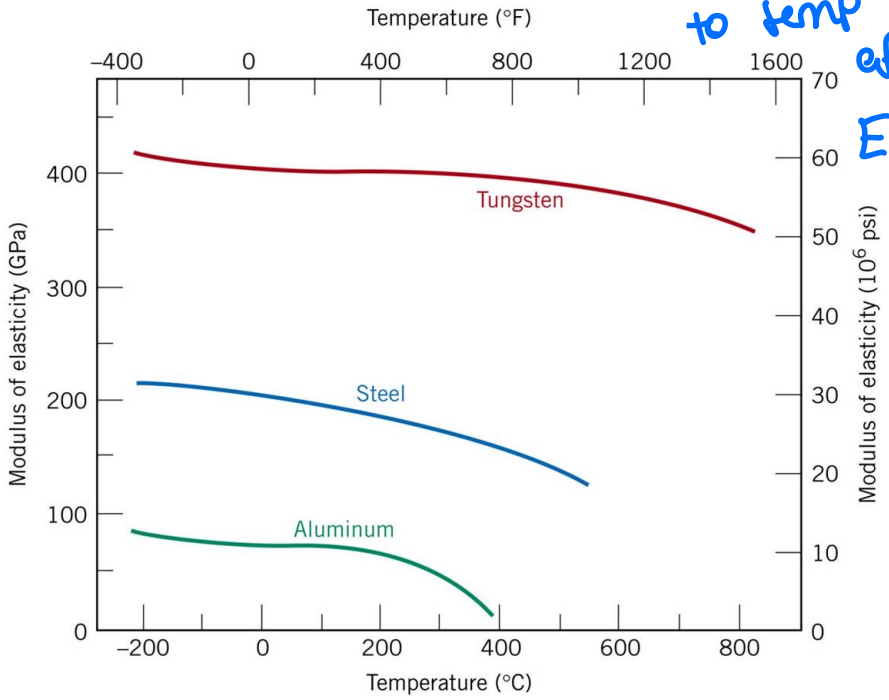
Low E value: - (not stiff)

Ceramics	300 GPa
Steel	207 GPa
Copper	110 GPa
Plastics	3 GPa

In general, the material's modulus (or stiffness) decreases with increasing temperature.

Can you think of why this happens?

because bonds become weaker



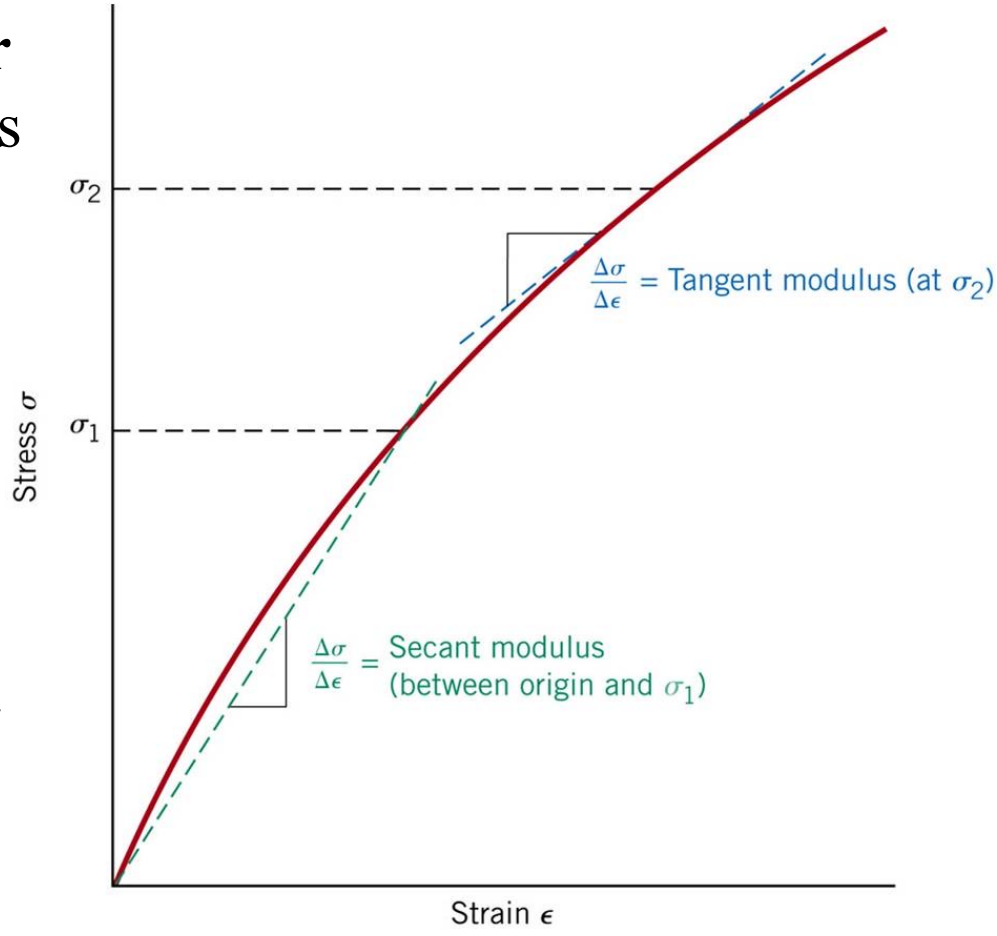
Temperature dependence of elastic moduli

Tangent and Secant Modulus

- Some materials do not show **linear** elastic region; their elastic region is non-linear. Cast iron, concrete, some polymers.

In this case E is harder to define:

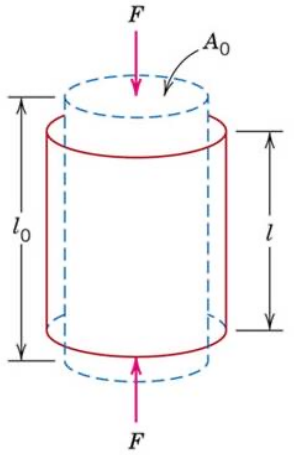
- We can use **Tangent modulus** which is slope of tangent at a particular stress level, or,
- **Secant modulus** which is slope of the line joining origin with some specified stress level.



Compression, Shear and Torsion Tests

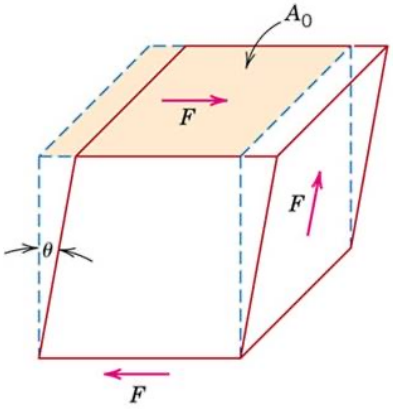
Compression

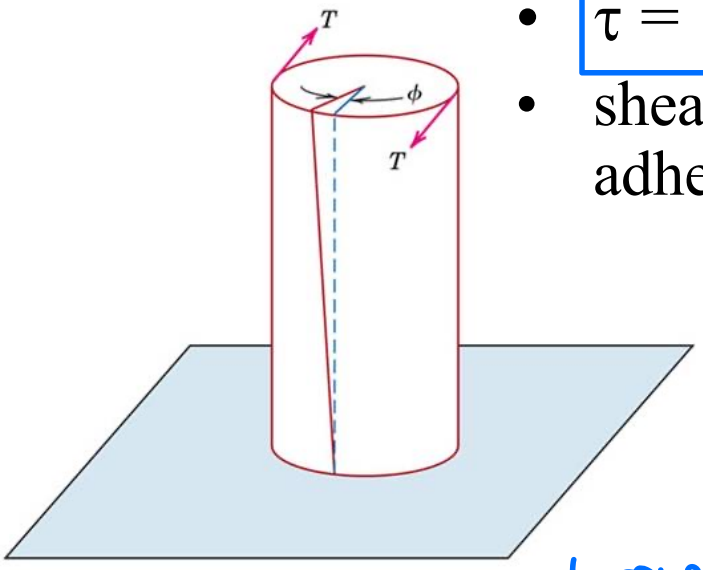
- by convention, stress and strain are negative
- used for measuring strength of brittle materials and for calculating forces required in manufacturing processes which involve compressive deformation



Shear

- shear stress is $\tau = F/A_o$ and γ (shear strain) is tangent of shear angle, θ
- $\tau = G\gamma$, G is shear modulus
- shear tests are often used to measure adhesive bonding, riveted joints etc





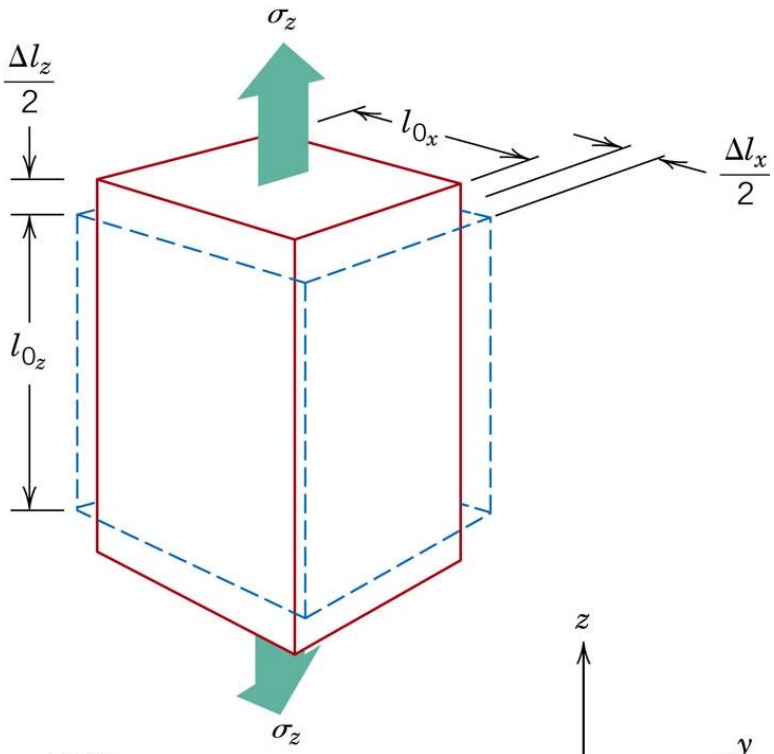
torque

Torsion

- *torsion* is a variation of shear occurring in *machine axles, drive shafts and twist drills*
- $T = f(\tau)$ and $\gamma = f(\phi)$

Poisson's Ratio

- When pulled in tension (along Z), a sample gets longer and thinner, i.e., *a contraction in the width (X) and breadth (Y)*
  - *if compressed gets fatter*



- **Poisson's ratio** defines how much strain occurs in the lateral directions (x & y) when strained in the (z) direction:

$$\upsilon = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$

can't be higher than 1

- Typical values = 0.2 to 0.5
- For isotropic materials

$$E = 2G(1+\nu)$$

Some materials are anisotropic so E & G vary with direction (e.g. composite materials and single crystals)

Elastic Deformation: Anelasticity

time dependence of elastic deformation

• So far we have assumed that elastic deformation is time ….independent.. (i.e. applied stress produces instantaneous elastic strain)

Some mats are time dependent

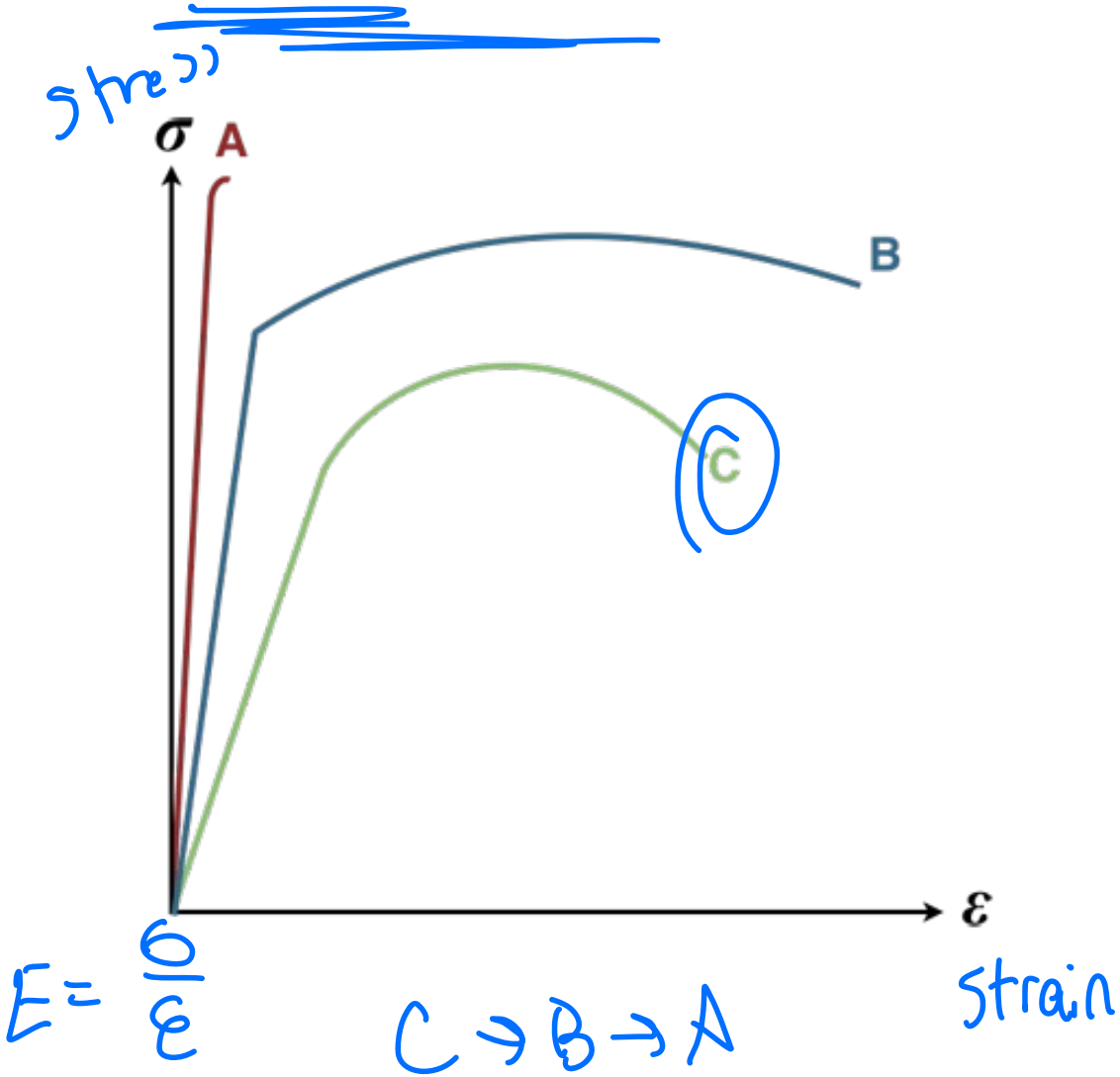
• In some materials, elastic strain may develop gradually after loading, and recovery may take time after unloading. This time-dependent, recoverable deformation is called **anelasticity**.

• The effect is normally small for **metals** but can be significant for **.**polymers**...** (“viscoelastic behavior”).

the load will be varied w/ time

Elastic Behaviour - *Review*

Which tensile engineering stress–strain curve represents the **least** stiff material?



Use the QR code shown in the classroom. This one won't work.



Next topic:

Plastic Deformation